NEGATIVE FREQUENCIES EXIST, BECAUSE THE DOT PRODUCT IS NEGATIVE FOR SAME DIRECTION, OPPOSITE HEADING

USEFUL MATH STUFF:

$$\sum_{n=N}^{M-1} \alpha^n \quad \frac{\alpha^N - \alpha^M}{1-\alpha} \qquad\qquad \sum_{n=N}^{M-1} a^n = \frac{a^N - a^M}{1-a}$$

$$\left|\sum f(x) g(x)\right| \leq \sum |f(x)||g(x)|$$

ENERGY $\quad E = \sum_{n=-\infty}^{+\infty} |x[n]|^2$

POWER $\quad P = \lim_{N \to \infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |s[n]|^2$

BOUNDED

s[n] IS BOUNDED IF

$$|s[n]| \leq B_s < \infty \;\; \forall n$$

ABSOLUTE SUMMABILITY

$$\sum_{n=-\infty}^{+\infty} |x[n]| < \infty$$

FOR AN LTI SYSTEM, IT GUARANTEES STABILITY

EVEN / ODD

$$S_{even}[n] = \frac{S[n] + S[-n]}{2} \qquad S_{odd}[n] = \frac{S[n] - S[-n]}{2}$$

SYSTEMS

MEMORY

MEMORY-LESS $\Leftrightarrow$ OUTPUT DEPENDS ONLY ON CURRENT VALUE OF INPUT

CAUSALITY

$h[n] = 0 \quad \forall n < 0$

STABILITY

BIBO

$$|x[n]| \le B_x < \infty \implies |y[n]| \le B_y < \infty \quad \forall n$$

IF LTI

$$\sum_{n=-\infty}^{+\infty} |h[n]| < \infty$$

PASSIVE

$$E_y \le E_x \longrightarrow \sum_{n=-\infty}^{+\infty} |y[n]|^2 \le \sum_{n=-\infty}^{+\infty} |x[n]|^2$$

RECURSIVE SYSTEMS

GENERAL EQUATION

$$\sum_{k=0}^{N} a_k[k]\, y[n-k] = \sum_{k=0}^{M} b_k[k]\, x[n-k]$$

LCCDE

$$y[n] = y_h[n] + y_p[n]$$

$$\sum_{k=0}^{N} a_k\, y[n-k] = 0$$

SIMILARITY MEASURES

CROSS - CORRELATION

$$C_{xy}[n] = \sum_{k=-\infty}^{+\infty} x[k]\, y[k-n]$$

AUTO - CORRELATION

$+\infty$

$$C_{xx}[n] = \sum_{k=-\infty} x[k]\, x[k-n]$$

$$C_{xx}[0] = E_x$$

DTFT

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n]\, e^{-j\omega n} \quad \longleftrightarrow \quad x[n] = \frac{1}{2\pi} \int_{\omega=-\pi}^{\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega$$

CONVERGENCE

ABSOLUTE SUMMABILITY $\displaystyle\sum_{n=-\infty}^{+\infty} |x(n)| < \infty$

UNIFORM CONVERGENCE $\displaystyle\lim_{M\to\infty} \left| X(e^{j\omega}) - \sum_{n=-M}^{M} x[n]\, e^{-j\omega n} \right| = 0$

DFT

SAMPLED DTFT

$$\sum_{n=-\infty}^{+\infty} x[n]\, e^{-j\omega n} \Big|_{\omega = \frac{2\pi}{N}k} = \sum_{n=-\infty}^{+\infty} x[n]\, e^{-j\frac{2\pi}{N}kn}$$

INVERSE

$$\tilde{x}[n] = \frac{1}{N} \sum_{n=-\infty}^{+\infty} X[k]\, e^{j\frac{2\pi}{N}kn}$$

MATRIX NOTATION

"$-$" IS INSIDE W_N

$$W_N := e^{-j\frac{2\pi}{N}} \longrightarrow W_N^{i} \Rightarrow \text{CLOCKWISE ROTATION IN STEPS OF } \frac{2\pi}{N}$$

$$X[k] = M\, x[n] \qquad x[n] = \frac{1}{N} M^{-1} X[k]$$

CIRCULAR NOTATION

$$x[n] \circledast_n h[n] \overset{DFT}{\Longleftrightarrow} X[k]\, H[k]$$

CIRCULAR CONVOLUTION MTX

$$\begin{matrix} h[0] & h[3] & h[2] & h[1] \\ h[1] & h[0] & h[3] & h[2] \\ h[2] & 1 & h[0] & 3 \end{matrix}$$

$h[3]$ 2 1 $h[0]$

OVERLAP ADD

1) DIVIDE INTO BLOCKS
2) PAD BLOCKS AND KERNEL TO N+M-1
3) DO CONVOLUTION
4) ADD AND COMPOSE OUTPUT

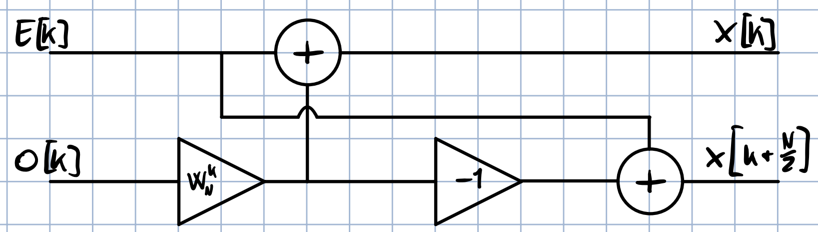
ADD: ADD ZEROS

OVERLAP SAVE

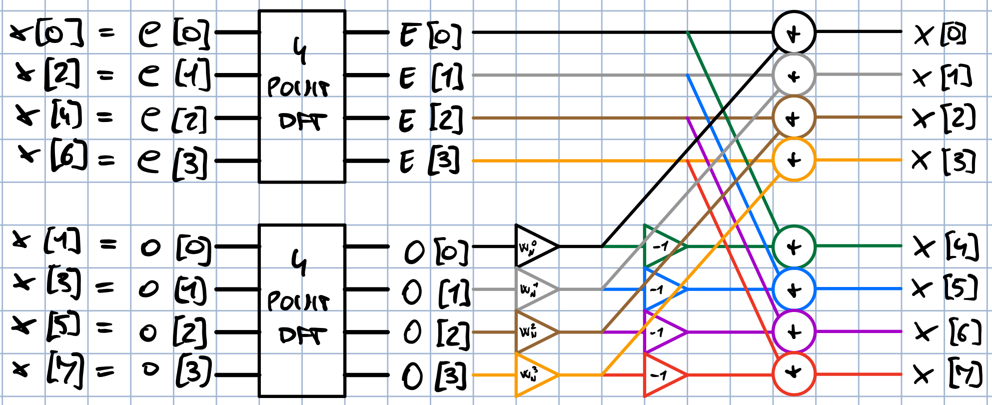
1) DIVIDE INTO BLOCKS
2) ADD ZEROS ON <u>LEFT</u> OF FIRST BLOCK TO REACH N+M-1
3) COPY M-1 VALUES ON LEFT OF BLOCKS
4) ADD ADDITIONAL BLOCK AT END AND PAD WITH ZEROS TO RIGHT
5) DO CONVOLUTION
6) EXCLUDE RESULTS OF PADDED VALUES
7) BUILD OUTPUT
8) CROP END

SAVE: SAVE PART OF NEXT BLOCK

BUTTERFLY FFT

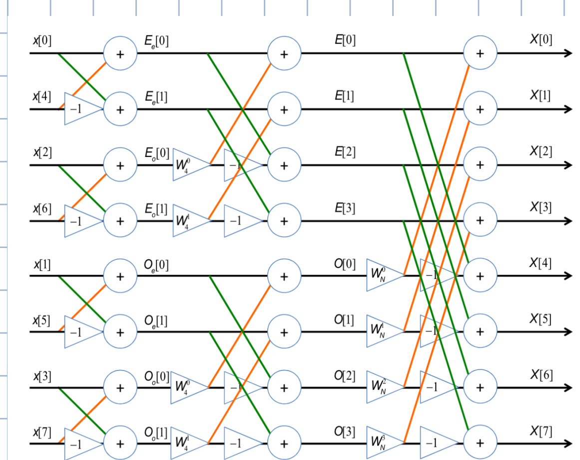


EVEN!



INDEX OF W_N^k IS WHAT IT GOES TO ON NEXT LAYER

ODD!



ODD SEQUENCES ALWAYS HAVE W_N^i

Z

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n] z^{-n} \quad , \; z := re^{j\omega}$$

ROC

$\lvert z \rvert > a$	CAUSAL
$\lvert z \rvert < a$	ANTICAUSAL
$a < \lvert z \rvert < b$	NON-CAUSAL / TWO-SIDED

KNOWN TRANSFORMS

$$X(z) = \frac{z}{z-a} = \frac{1}{1-az^{-1}} \longleftrightarrow a^n u[n] \;\Big/\; -a^n u[-n-1]$$

$$X(z) = 1 \longleftrightarrow x[n] = \delta[n] \qquad X(z) = z^{-a} \longleftrightarrow x[n] = \delta[n-a]$$

ALL PASS DECOMPOSITION

$$H(z) \cdot \text{ALLPASS} = H(z) \frac{z^{-1} - \alpha^*}{1 - \alpha z^{-1}} = H(z)$$

SYSTEMS IN Z

STABILITY

$$r = 1 \in \text{ROC}$$

LCCDE

USE JUST COEFFICIENTS TO BUILD $H(z)$

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k} \quad \leftarrow \text{COEFFS. OF } x}{\sum_{k=0}^{N} a_k z^{-k} \quad \leftarrow \text{COEFFS. OF } Y}$$

DISTANCE MAGIC

$$|H(e^{j\omega})| = \frac{\text{PRODUCT OF DISTANCES OF ZEROS FROM } \omega}{\text{PRODUCT OF DISTANCES OF POLES FROM } \omega}$$

ALL-PASS

$$H(z) = \frac{z^{-1} - P_k^*}{1 - P_k z^{-1}} = \frac{1 - P_k^* z}{z - P_k}$$

MINIMUM PHASE + ALL-PASS DECOMPOSITION

CONCEPT: ISOLATE CRITICAL POLE AND ARTIFICIALLY EXTRACT ALL-PASS FORMULATION → RESULT IS MINIMUM PHASE · ALL-PASS

FIR

	ODD LENGTH	EVEN LENGTH
SYMM	TYPE 1 ALL FILTERS	TYPE 2 LPF + BPF
ANTISYMM	TYPE 3 BPF	TYPE 4 HPF + BPF

LENGTH

IIR

$$\omega_p = 2\pi \frac{f_p}{f_0} \quad , \quad \omega_s = 2\pi \frac{f_s}{f_0}$$

$$|H(j\Omega)|^2 = \frac{1}{1+\left(\frac{\Omega}{\Omega_c}\right)^{2N}} \longrightarrow \text{POLES SPACING } \frac{\pi}{N}$$

$\downarrow$

CHOOSE ONLY $Re\{p\} < \phi$

$$N \geq \frac{1}{2} \frac{\log_{10}\left(\frac{A^2-1}{\varepsilon^2}\right)}{\log_{10}\left(\frac{\Omega_s}{\Omega_p}\right)}$$

$$A_p = \frac{1}{1+\varepsilon^2}$$

$$A_s = \frac{1}{A^2}$$

BILINEAR

$$s = \frac{2}{T} \frac{z-1}{z+1}$$

$$\Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right)$$